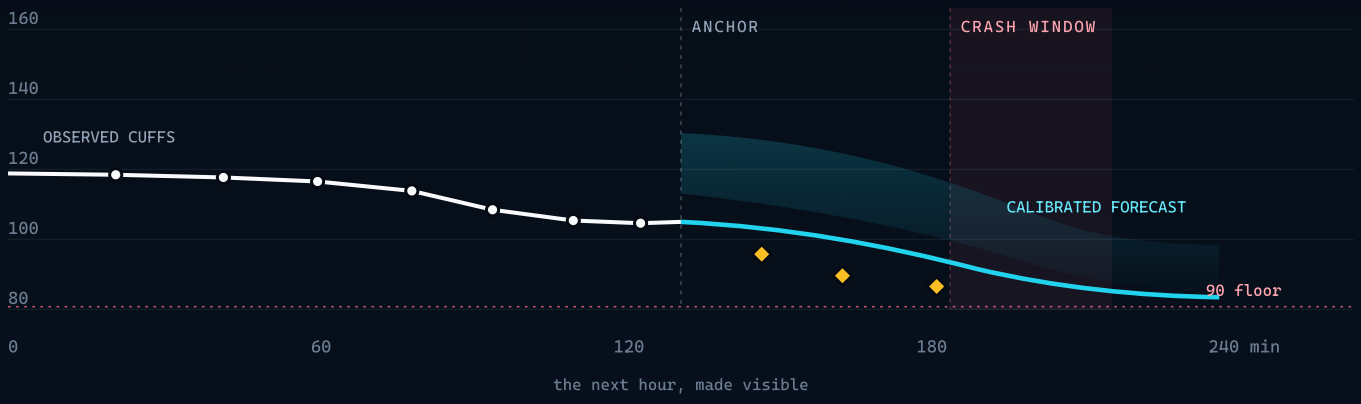
See the pressure falling. Show the trade-off.

A chair-side session twin for haemodialysis: a calibrated 60-minute forecast, exact fluid arithmetic, and a structured review pathway — for the technician at the chair, not a prescription engine.



01 · SEE

Where pressure is heading in the next 60 minutes — with a calibrated band, not a bare number.

02 · STRUCTURE

What the unit's deterministic protocol says next, with role-scoped review steps.

03 · PRICE

What each fluid option leaves behind: fluid left, next-session UF, and the trade-off.

INSIDE THIS OVERVIEW

02 The chair-side gap
03 The product
04 The evidence
05 Seven pre-registered checks
06 The decisive negative
07 The app
08 Why it matters
09 The boundary
10 Judge Q&A

> “The gap is not another score.
> It is time and a shared trade-off.”

HEMOTWIN V2 · PRODUCT THESIS

205,482 SESSIONS · 1,070 PATIENTS · FOLD 2 UNTOUCHED · G7 TWIN TEST: FAIL — REPORTED, NOT HIDDEN

Research prototype

NOT A MEDICAL DEVICE
NO TREATMENT RECOMMENDATION

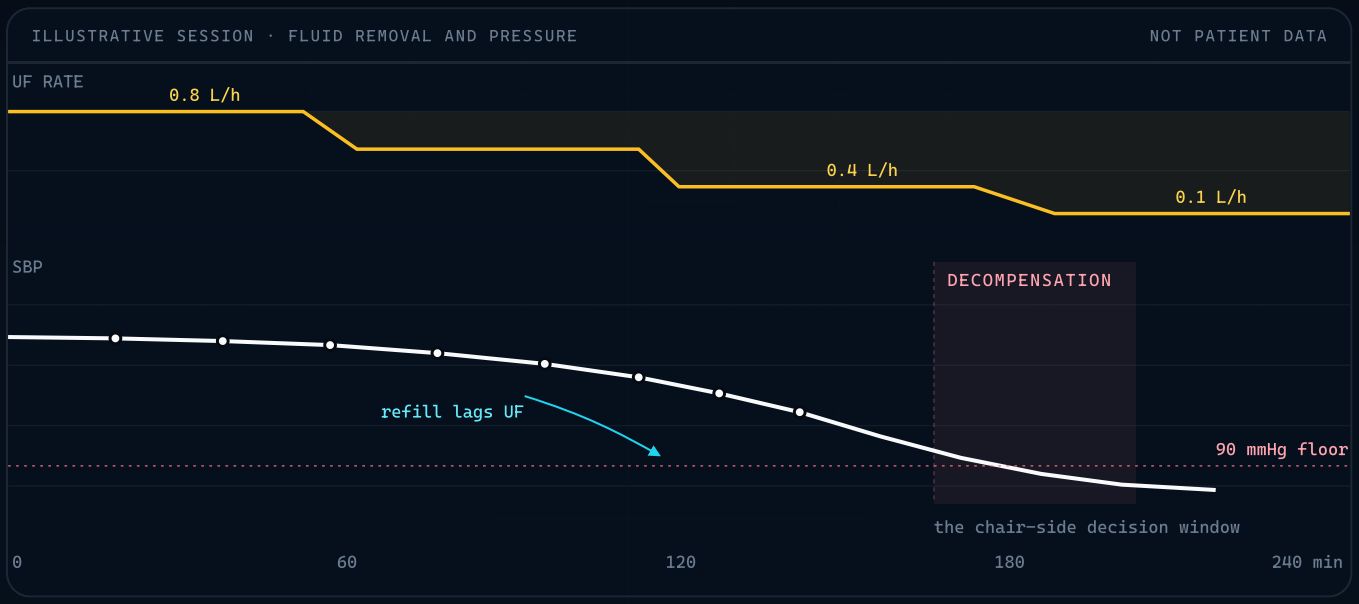
10.99	89.2 / 90.1%	581	0
FOLD-2 MAE · MMHG	BAND COVERAGE	MATCHED G7 EVENTS	OPERATIONAL UF GAIN
Fair 59-feature model	HOLD / TAPER	158 patients · pre-registered	the twin is an illustration

01 · THE CHAIR-SIDE GAP

The dangerous minutes are the ones nobody is looking at.

During dialysis, fluid is removed faster than the body can refill the bloodstream. When plasma refill lags ultrafiltration, pressure falls — sometimes before anyone at the chair has a shared plan.

Haemodialysis filters blood outside the body. Blood leaves through vascular access, passes through a dialyzer, and returns. **Ultrafiltration (UF)** removes the fluid gained between sessions. **Plasma refill** moves fluid from tissue back into the vessels. If UF outruns refill, intravascular volume falls; the baroreflex compensates, then exhausts. The result is intradialytic hypotension (IDH) — here defined as a systolic pressure below 90 mmHg — often with symptoms, sometimes with a crash.



Who is at the chair?

Often a dialysis technician or nurse, while the nephrologist is remote. The next action — reduce UF, pause, give fluid, call for advice — happens before the doctor arrives. The missing piece is not another score. It is **time** and a **shared trade-off**.

> The model can rank risk. It cannot make the response happen. That is why the product is a workflow layer, not a prediction dashboard.

Cuff cadence is often around 30 minutes. The warning window is 15–60 minutes. The session can change between two readings — which is why a projection matters.

10.4%
HEMOBP SESSIONS
labelled IDH, nadir <90 mmHg

25.5%
SPANISH COHORT
clinician-defined symptomatic label — a different definition, never combined

34.5%
OF THE COHORT
diabetic patients

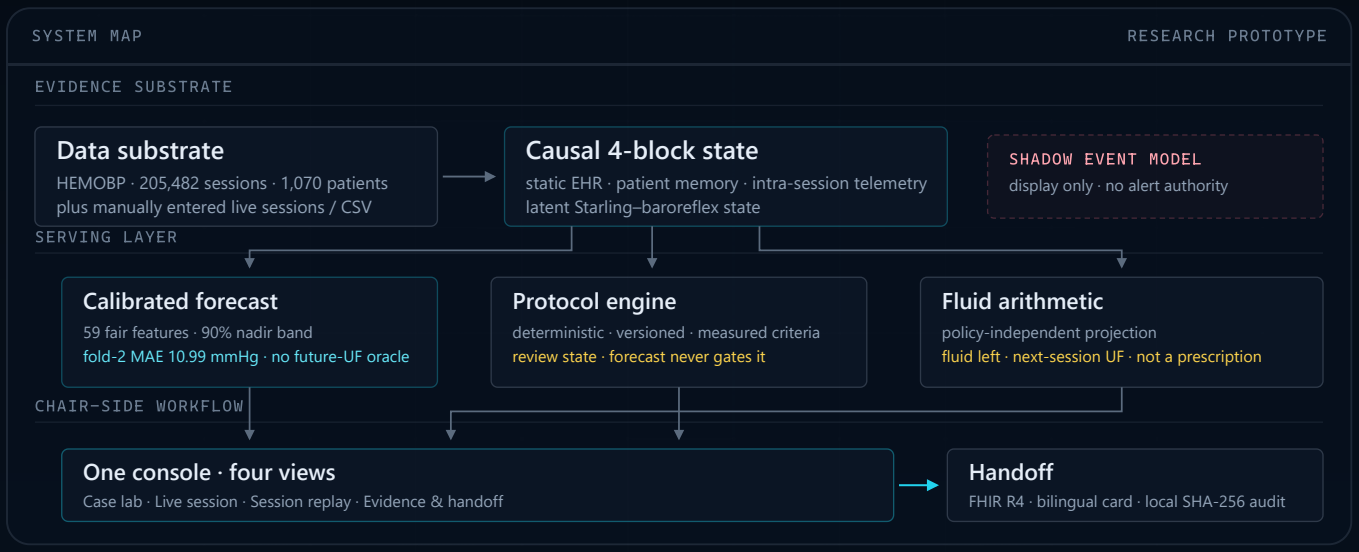
8.2%
EVENT MODEL DETECTION
at the pre-registered alert budget — below the current-SBP rule at 20.2%

> “The gap is not another score. It is time and a shared trade-off.”
>
> HEMOTWIN V2 · PRODUCT THESIS

02 · THE PRODUCT

One console. Four views. One shared state.

A device-agnostic session-state mirror. It does not control the machine. It makes the next review legible: where pressure is heading, what the unit's protocol says, and what each fluid option leaves behind.



The four views

CASE LAB	**Read the projection.** Seven curated sessions; click a cuff for SBP/DBP/UF and the twin state at that minute.
LIVE SESSION	**Mirror a real session.** Manual/CSV entry, stale-cuff guardrail, protocol state, arithmetic, local audit.
REPLAY	**Watch time move.** The forecast draws forward; recorded cuffs and UF steps arrive at their real times.
EVIDENCE	**Inspect the boundary.** Gates, G7, claim boundary, FHIR, bilingual handoff, audit chain.

What it is — and is not

Mirrors state	No machine control
Structures review	No prescription
Prices arithmetic	No effect claim
Builds a handoff	Not immutable

Runtime boundary

No authentication, no role enforcement, no machine integration, no external notification, no network call at runtime.

The role selector is a recorded label, not enforcement. The audit chain is a local demo object.

> "Mirror the session. Structure the response. Show the fluid arithmetic."
>
> HEMOTWIN V2 · ONE-LINE PRODUCT DEFINITION

01	02	03	04	05
Cuff	**Mirror**	**Forecast + protocol**	**Fluid arithmetic**	**Handoff**
Recorded at the chair	One shared session state	The next hour and the next step	What each option leaves behind	FHIR · bilingual · audit

SBP / DBP	UF	LATENT	FORECAST	PROTOCOL	FLUID
Current cuff + history	Rate + cumulative volume	Deficit · refill · baro	+60 min mean + band	Review state + steps	Left · next-session UF

03 · THE EVIDENCE

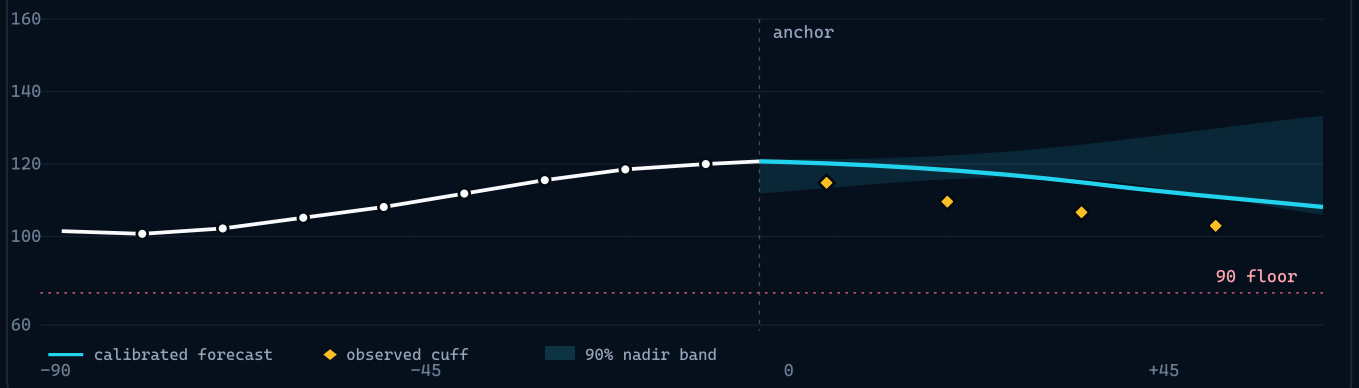
A calibrated forecast, not a crystal ball.

The console ships one forecast: a fair, pre-anchor model that never sees future ultrafiltration. A second, oracle-assisted number exists only as a display comparison — it is not the baseline.

205,482	59	10.99	89.2 / 90.1%
QC-ELIGIBLE SESSIONS	FAIR FEATURES	FOLD-2 MAE · MMHG	90% BAND COVERAGE
HEMOBP · 1,070 patients · 4.19M rows	static EHR + memory + telemetry	RMSE 14.65 · no future-UF oracle	HOLD / TAPER · marginal, not conditional

WHAT THE CONSOLE SHOWS · ONE CURATED SESSION

ILLUSTRATIVE · NOT THE PATIENT RECORD



THE CONSOLE FORECAST

Fair pre-anchor model

59 features. No future-UF oracle block. Fold-2 MAE **10.99**, RMSE 14.65. This is the calibrated floor the app uses.

DISPLAY COMPARISON ONLY

Oracle-assisted forecast

72 features including 13 observed future-UF features. Fold-2 MAE **9.60**. A smaller number, not a better twin — and not the G2 baseline.

The band is a 90% **nadir interval**, widened along the path for display. It is not a per-minute confidence band. Coverage is marginal, not conditional.

What the data is

HEMOBP, Taiwan: de-identified, CC BY 4.0. 205,482 QC-eligible sessions, 1,070 patients, 4,190,926 telemetry rows. The combined raw scale is 303,122 sessions across Taiwan and a Spanish frame. The Spanish transport check failed: **AUROC 0.474** vs 0.720 internally. The labels mean different things.

Two labels, never combined

10.4%	25.5%
HEMOBP	SPANISH FRAME
nadir SBP < 90 mmHg	clinician-defined symptomatic hypotension

No external-validity claim. The method travels; the weights do not.

STEP 1

Anchor state

Everything strictly at or before the anchor: static EHR, prior sessions, current cuff, UF, machine settings.

STEP 2

59 fair features

No future UF, no future cuffs, no outcome. The feature blocks are frozen before serving.

STEP 3

Calibrated mean

The 60-minute nadir estimate is calibrated on held-out patients; fold 2 is untouched.

STEP 4

90% band

The band is a nadir interval, widened along the display path. Marginal coverage; not a guarantee.

04 · WHAT WE TESTED

Seven checks written before we looked.

G1–G7 are pre-registered gates, not a score. Five of the six G1–G6 booleans pass; G3 fails. G7 also fails. The failures are part of the result, not an appendix.

GATE	WHY WE TESTED IT	OUTCOME	WHAT IT MEANS
G1 · Taper	Is the new pre-anchor forecast better than the retired v3 twin?	`11.59 vs 15.55` MAE PASS 374 taper anchors	The pre-anchor substrate is better. Not because of the ODE.
G2 · Baseline	Is it honest against its own baseline and against the oracle display?	`10.99 / 9.60` PARTIAL pass vs own base; fail vs oracle	At gain 0 the baseline is a tie — not evidence of a treatment effect. The oracle is a display comparison.
G3 · Actual vs hold	Does the twin predict an actual treatment effect, or is it identical to holding UF?	`0.000` difference FAIL 397 rows	Actual and hold predictions are identical at calibrated gain 0. The what-if gate fails.
G4 · Monotonicity	Does the policy model behave consistently and continuously?	`0` violations PASS 3,086,844 checks	Engineering consistency, not efficacy.
G5 · Coverage	Do the 90% uncertainty bands cover reality?	`89.2 / 90.1%` PASS HOLD / TAPER	Marginal coverage, not a conditional guarantee.
G6 · Invariance	Does it stay bounded for new patients and ignore future data?	`200 / 200` PASS future-invariant · 284 cold starts	Leakage control and a cold-start guardrail.
G7 · UF effect	Does reducing UF early actually raise SBP?	`+0.13` mmHg per 0.25 L/h FAIL CI [−2.63, +3.06]	No positive effect. The twin is a mechanism illustration; operational gain 0.

How to read a pass. A pass is an engineering check against a pre-registered criterion. It is not clinical validation. G2 is a tie at gain 0, not evidence of a treatment effect.

What the failures retire. Quantitative treatment recommendations, the actual-UF physiological twin, and the event model's alert authority. The console keeps the forecast, the protocol, and the arithmetic.

Also reported, not hidden

- **Task 01 twin:** taper MAE gain 4.9% vs 10% required; 21.2% worse than a direct GBDT under the oracle. Retired.
- **Task 02 alert utility:** AUROC 0.880, but 8.2% detection at the 0.26 false-alert budget vs 20.2% for the current-SBP rule. Shadow only.
- **Twin-v3 memory:** 13 patient-memory features added +0.05% on dev. Pre-registered gain gate failed; fold 2 untouched.

Negative controls that matter

- **Sham-time control:** +2.91 mmHg [2.31, 3.54]. It fails — the matched design is not clean enough to support a positive claim.
- **P6a trajectory:** +0.43 mmHg [0.15, 0.65] worse out of sample. The twin does not improve the forecast.
- **Spanish transport:** AUROC 0.474 vs 0.720 internal. Labels differ; no external-validity claim.

228 package tests · 0 failures **14/14** claim-boundary checks **Package verifier** all checks pass

Fold 2 never used for tuning

05 · THE DECISIVE NEGATIVE

We tested the twin. It did not work.

If reducing UF early raises SBP, the twin has a treatment effect. We froze the protocol before estimation and tested it on 581 matched events in 158 patients.

Design

- Non-reactive UF step-down
- Matched within-patient controls
- Continuous dose-response
- Pre-anchor rule
- Sham-time negative control
- 581 matched events / 158 patients
- Fold 2 untouched

PRIMARY ESTIMATE

+0.13

mmHg per 0.25 L/h · 95% CI [−2.63, +3.06]

Indistinguishable from zero. The interval includes both a meaningful benefit and a meaningful harm.

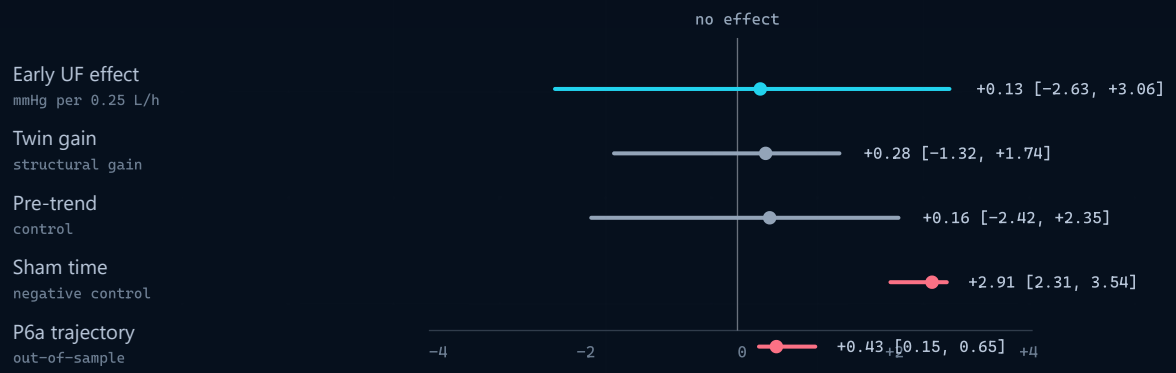
PRE-REGISTERED

Protocol before results

Frozen at SHA-256 `9ec625…fd73` before estimation. 581 matched events across 158 dev patients. Fold 2 held back.

VERDICT · FAIL

G7 ESTIMATES · 95% BOOTSTRAP CI — PRE-REGISTERED · NOT A TREATMENT CLAIM



VERDICT · FAIL

The effect is indistinguishable from zero. The pre-registered sham-time control fails. P6a gets worse out of sample. The operational gain stays **0**.

The twin remains a **labelled mechanism illustration**. The dashed fan is off by default. The calibrated forecast and the fluid arithmetic do not depend on the twin's treatment effect.

Why this page exists. A product that claims a treatment effect it cannot identify is not decision support. The negative result is the reason the rest of the console can be trusted.

What we keep. The calibrated forecast, the deterministic protocol, the fluid arithmetic, and the handoff. None of them depends on a UF-effect claim.

What we retired. The quantitative treatment recommendations, the actual-UF twin, the alert authority, and the personalized alert channel. The mechanism sketch stays labelled and off by default.

STAGE 0	FOLD 2	POWER	PROTOCOL
581 matched events across 158 dev patients	65 pre-anchor events · 28 at anchor	Anchor fan underpowered: 28 < 50 events	Frozen at SHA-256 9ec625…fd73 before estimation

06 · THE APP

The system, in the room.

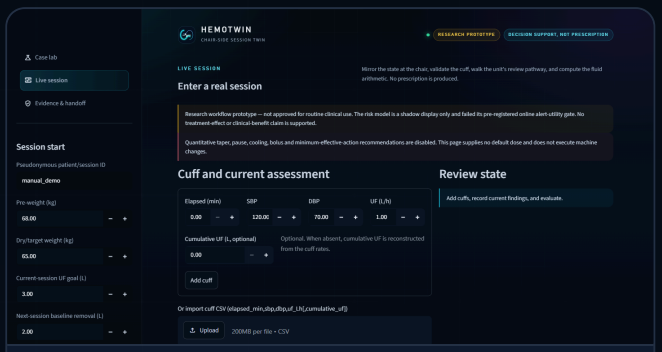
One Streamlit console. Every view reads the same session state. The screenshots below are the shipped app, not mockups.



Case lab

Read the forecast. Click a cuff for SBP, DBP, UF and the twin state at that minute.

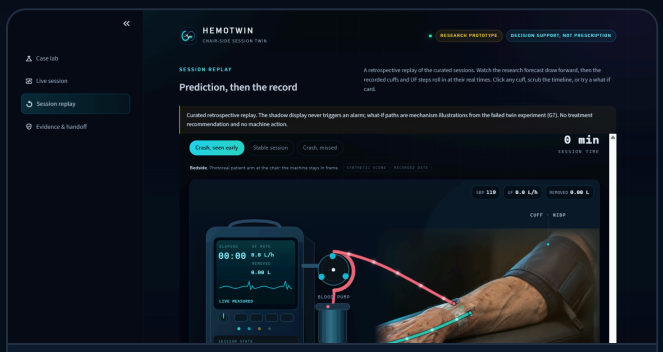
01



Live session

Mirror a real session; the stale-cuff guardrail refuses to guess; protocol state and arithmetic update.

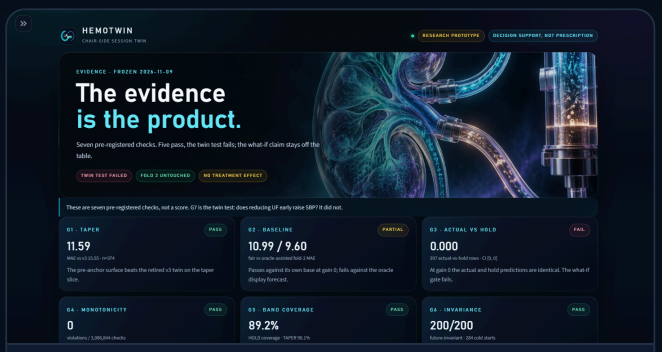
02



Session replay

Watch the forecast draw forward, then the recorded cuffs and UF steps arrive at their real times.

03



Evidence & handoff

Gates, G7, claim boundary, FHIR, bilingual handoff, audit chain, 12-chair pod triage.

04

Component → view map

Calibrated forecast	Case lab · Live session · Replay
Protocol engine	Case lab · Live session
Fluid arithmetic	Case lab · Live session
Shadow event score	Display only · no alert authority
FHIR / bilingual / audit	Evidence & handoff

What each view is for

Case lab is the teaching surface: seven curated stories, the fan chart, click inspection, and the fluid table. **Live session** is the working surface: manual entry, validation, protocol, arithmetic, audit. **Replay** is the temporal surface: it makes "forecast, then the record" visible. **Evidence** is the trust surface: it shows what passed, what failed, and what we refuse to claim.

> The app has no authentication, no role enforcement, no machine integration, and no network call at runtime.

228 package tests · 0 failures **14/14** claim-boundary checks **Package verifier** all checks pass

0 console errors in the shipped app

07 · WHY IT MATTERS

It does not replace the nurse. It gives the call a shared state.

The product's real difference is not a better number on a dashboard. It is a shared, inspectable session state that lets the chair-side team act earlier and ask for help with the trade-off already on the table.

THE WORKFLOW DIFFERENCE — ILLUSTRATIVE

TODAY

Notice the drop	Call the nephrologist	Hold a verbal trade-off	Decide under pressure
a cuff, a symptom, a feeling	describe the session by voice	fluid left? next session?	document in notes / phone
reactive	remote	in someone's head	not shared

WITH HEMOTWIN

Mirror the session	Forecast + protocol	Price the fluid	Hand off
cuffs · UF · protocol state	next 60 min · review step	left behind · next-session UF	FHIR · bilingual · audit
same state	shared	arithmetic	inspectable

The decision remains human. The difference is that the trade-off arrives with a shared, inspectable state.

Earlier attention

Makes the next 60 minutes visible while the patient is still stable enough to review — not only after a cuff is already low.

Explicit trade-off

Each option carries a fluid cost and a next-session UF intensity. The conversation stops living in someone's head.

Shared handoff

Protocol step, role, FHIR bundle, bilingual shift card, local audit chain. The remote doctor sees the same state.

What “real difference” means here

The difference is not that the model is always right. The forecast is a calibrated **conditional 60-minute estimate**; the 90% band is marginal, not a guarantee. The difference is that the chair-side team can see the projection, the protocol step, and the fluid consequence **together**, and hand that state to a remote clinician without reconstructing it from memory.

Not yet proven. Workflow benefit has not been prospectively validated. The next experiment is a silent-mode pilot, then a pre-registered workflow evaluation — not another model. The product must earn that claim.

PILOT MEASURE	PILOT MEASURE	PILOT MEASURE	PILOT MEASURE
Time from trigger to documented review	Share of sessions with an explicit fluid trade-off	Escalation rate and nurse-reported usability	Fluid carried forward; next-session UF intensity

What would count as a win

Faster documented review, with the same clinical team and no extra alert burden.

What would count as a miss

No measurable time gain, or a trade-off that is shown but never used in the record.

What we will not do

Claim a workflow benefit before the pilot. The product must earn that result.

08 · THE BOUNDARY

The boundary is part of the product.

The console says what it can and cannot do. That is not a disclaimer at the end; it is the design constraint that keeps the rest honest.

WE CAN CLAIM

- A calibrated pre-anchor nadir forecast: fold-2 MAE **10.99**, RMSE 14.65.
- A 90% nadir band with marginal coverage: **89.2% HOLD / 90.1% TAPER**.
- Exact arithmetic fluid carry-forward for user-selected scenarios.
- A deterministic, versioned review pathway with role-scoped steps.
- A local FHIR R4 export, bilingual handoff card, and SHA-256 demo audit chain.
- Honest negative results from the twin, event model, and external transport tests.

WE REFUSE TO CLAIM

- A validated counterfactual or treatment effect from the twin.
- That any UF policy will raise SBP, prevent IDH, or is minimum effective.
- Alert utility, clinical benefit, or a safety layer.
- Prospective, external, symptomatic-IDH, or Indian-unit validation.
- A prescription, a default dose, or autonomous machine action.
- Authentication, role enforcement, immutable audit, or machine integration.

Next evidence we owe

1 · Silent mode	2–4 weeks beside the team. No machine control, no alert authority. Log what the console would have shown.
2 · Local labels	Harmonise the IDH definition, symptoms, and nursing actions; recalibrate on partner-machine data.
3 · Workflow trial	Pre-registered prospective evaluation of time-to-review, escalation, and fluid carry-forward.
4 · India	No Indian cohort yet. The method travels; the weights do not. Retraining and local recalibration are prerequisites.

Why the boundary earns trust

A system that knows what it cannot do is easier to audit, easier to pilot, and harder to misuse. The G7 failure is not hidden in an appendix: it is on the evidence page and in the product copy. The twin is an illustration; the forecast and the arithmetic are the validated surfaces.

> **Deployment context.** India has roughly 13,800 dialysis machines across 751 PMNDP districts and 1–1.5 nephrologists per million people. The cohort is 34.5% diabetic. Those numbers describe the need, not an efficacy claim.

What would change our mind

Local recalibration changing the forecast, or a workflow trial showing no time benefit.

What would stop the pilot

An alert burden that outweighs the value, or a trade-off that cannot be used at the chair.

What we will report

The same numbers either way: time-to-review, escalation, fluid carry-forward, and nurse usability.

> “The twin knows what it cannot do; that is what makes it usable as a chair-side decision-support layer.”
>
> HEMOTWIN V2 · CLOSING POSITION

09 · JUDGE Q&A

The questions we expect — answered plainly.

Answer, then stop talking. The evidence page has the numbers; this page has the short form.

What is HemoTwin?

One console with four views: Case lab, Live session, Session replay, and Evidence & handoff. It mirrors the session, structures the unit's review pathway, and makes the fluid arithmetic explicit. It does not prescribe.

Why not just predict IDH?

Because ranking is not timing. Our event model ranked at AUROC 0.880 but detected only 8.2% of event sessions at the pre-registered alert budget — below the current-SBP rule at 20.2%. The product is the workflow layer; the model stays shadow.

Did the physiological twin work?

No. G3 is identical at gain 0. G7 gives +0.13 mmHg per 0.25 L/h with a 95% CI of [−2.63, +3.06]; the sham control fails; P6a gets worse. Verdict FAIL. The fan is a labelled illustration and is off by default.

Which forecast is which?

The console uses the fair 59-feature pre-anchor model: MAE 10.99, no future-UF oracle. The 9.60 is an oracle-assisted display forecast with 13 observed future-UF features. It is not the G2 baseline and not a better twin.

Is this a medical device?

No. It is a research workflow prototype. It does not diagnose, prescribe, control the machine, or claim clinical benefit. The role selector is a recorded label, not enforcement.

Is the audit trail real?

It is a local SHA-256 chain over demo events. It is append-only by construction, but not authenticated, not tamper-evident, and not an immutable clinical record. The app says so.

You have no Indian data.

Correct. HEMOBP is Taiwanese; the Spanish transport check failed at AUROC 0.474 and the labels differ. No Indian cohort has been collected. The method travels; the weights do not. Local retraining and recalibration are prerequisites.

What is the evidence?

205,482 QC-eligible HEMOBP sessions, 1,070 patients, 4,190,926 telemetry rows, CC BY 4.0. G7 uses 205,113 sessions; 369 lack the required pre-anchor record. Labelled hypotension is 10.4% (nadir <90); the Spanish symptomatic label is 25.5% — different labels, never combined. Fold 2 was untouched.

What would a pilot look like?

A silent-mode run beside the team for 2–4 weeks: no machine control, no alert authority, log what the console would have shown. Then local recalibration, then a pre-registered prospective evaluation of time-to-review and workflow. The next experiment is a workflow, not another model.

What do you refuse to claim?

A validated counterfactual or treatment effect from the twin; that any UF policy will raise SBP, prevent IDH, or is minimum effective; alert utility, clinical benefit, or a safety layer; prospective, external, symptomatic-IDH, or Indian-unit validation; a prescription, default dose, or autonomous machine action.

Three-minute demo path

1. **Case lab** — hover a cuff for calibrated + observed SBP; click for SBP, DBP, UF and the twin state.
2. **Fluid arithmetic** — the projection is shared; the table shows what changes.
3. **Evidence** — open G7, the claim boundary, the FHIR bundle, the audit chain.
4. **Replay / Live** — watch time move; enter a session and evaluate.

References

Lin et al., HEMOBP, *Sci Data* 2019. Mendeley Dialysis Database, Spain. Zhang et al., *NDT* 2023. Tsujimoto et al., Cochrane 2019. MyTEMP, *Lancet* 2022. CLIMB, *JAMA* 2005 / *KI* 2020. Mitsides et al. 2019. Stefánsson et al., *KI* 2014. KDOQI 2015. Full citations and claim boundary: `v2/docs/FACTS.md`, `v2/docs/EVIDENCE_STATUS.md`.

`4 views  228 package tests  14/14 claim-boundary checks  0 network calls at runtime  Fold 2 untouched`

> “The twin knows what it cannot do. That is what makes it usable.”
>
> HEMOTWIN V2 · RESEARCH PROTOTYPE